

A role of steam activation on CO₂ capture and separation of narrow microporous carbons produced from cellulose fibers



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ABSTRACT

Ultra-microporous carbons were prepared from CF (cellulose fibers) by physical activation at various temperatures. Physical activation using steam was observed to have an influence on the development of new pores and the expansion of pore sizes and to be effective in developing optimal micropores for CO₂ adsorption on the carbon surface. On optimizing the preparation conditions, the synthesized materials exhibited a high CO₂ adsorption capacity of 3.776 mmol g^{−1} at 298 K and 1 bar, as well as an impressive CO₂/N₂ selectivity of 47.1. The prepared samples show excellent regeneration property throughout 10 adsorption–desorption cycles. These results demonstrate the successful fabrication of porous carbons with enhanced CO₂ adsorption capacities and CO₂/N₂ separation abilities without chemical activation.

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1. Introduction

In recent years, public awareness of the global climate change crisis has increased and methods to reduce emissions of greenhouse gases such as carbon dioxide (CO₂) are being intensively researched [1–4]. Several strategies for reducing CO₂ emissions have been proposed, and studies have been directed toward CCS (CO₂ capture and sequestration) [5]. A number of separation technologies could be applied with post-combustion step. These include adsorption, absorption, membranes, and cryogenics. Adsorption method uses porous solid CO₂ adsorbents such as zeolites, porous silicas, MOFs (metal organic frameworks), and porous carbons. These adsorbents are regenerated by TSA (Thermal swing adsorption) or PSA (pressure swing adsorption) [6–9]. Zeolites have a high adsorption capacity under mild conditions; however, this capacity decreases in the presence of impurities [10]. Porous silicas (such as MCM-41 and SBA-15) have no strong interactions with CO₂ because such interactions cannot be induced by their residual hydroxyl groups [11]. MOFs are expensive and their adsorption–desorption performance diminishes over multiple cycles [12–14]. Porous carbons have several advantages over other porous materials. Firstly, porous carbons have various properties

that make them suitable for CO₂ capture; e.g., rapid adsorption kinetics, a high specific surface area, and hydrophobicity [15–19]. Secondly, they can be prepared from a wide range of inexpensive starting materials, which include coal and biomass polymers such as cellulose [20–22]. Thirdly, CO₂ desorption from porous carbons can be easily carried out using the pressure swing approach; their regeneration, therefore, has low energy requirements [23–25]. In this work, CF (cellulose fibers) were chosen as the starting material in the preparation of porous carbons. Cellulose fibers are the most renewable and abundant biomass polymers; they are the constituents of wood, cotton, and other plant-based materials. They are, therefore, economically viable, eco-friendly, and suitable porous carbon precursors [26–28]. In recent years, studies on carbon-based adsorbents have attempted to increase the surface area and fine-tune the pore structure via activation processes. There are two types of activation processes, chemical and physical, that have been reported as methods for increasing the surface area of porous carbons [29–31]. There is no doubt that chemical-activation processes have a superior ability to enhance the specific surface area and create highly developed porous structures. However, chemical activation has certain disadvantages. Firstly, a large amount of chemical agents are consumed during chemical activation, generating wastewater and causing environmental pollution. Secondly, chemical activation is relatively complex. Thirdly, this process results in higher costs, including maintenance charges for corroded equipment [32–34]. Physical activation such as with steam is a

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good alternative. Activation by steam produces a wider pore-size distribution, with better development of wide micropores and mesopores than activation by CO_2 , which is generally known to result in narrow microporosity. Moreover, physical activation is less expensive and more ecofriendly than chemical activation [35–37].

In this study, we successfully controlled steam activation to produce structurally well-ordered porous carbons for CO_2 capture. Porous carbons were prepared from cellulose fibers by physical activation with steam, and their characteristics such as specific surface area, pore-size distribution, and pore volume were investigated. The CO_2 adsorption capacities were measured at 273 K and 298 K. The heat of adsorption (ΔH_{ads}) of CO_2 was calculated from CO_2 adsorption isotherms and the CO_2/N_2 selectivity was calculated from CO_2/N_2 adsorption isotherms at 298 K. Our results indicate that steam-activated porous carbons are promising adsorbents for CO_2 capture and CO_2/N_2 separation.

2. Experimental

2.1. Materials and sample preparation

Cellulose fibers (medium) were purchased from Sigma–Aldrich and used as the starting material in the preparation of high specific surface area porous carbons.

These porous carbons were synthesized using a simplified method as follows: Cellulose fibers were placed in a tubular furnace and treated at 200 °C under a flowing N_2 atmosphere (200 cc/min) for 60 min (pre-pyrolysis). The dried cellulose fibers were then heated to a target temperature in the range of 700–850 °C. When the furnace temperature reached this target temperature, carbonization was carried out for 30 min. The carbonized materials were then physically activated with steam (5 ml/h) for 60 min. To maximize the activation effect for activation process, the gauge pressure in the tubular furnace was maintained at 1.2 bar. After activation, the sample was cooled to room temperature. The obtained samples were washed by ethanol, followed by with deionized water. The CF (cellulose fiber) and steam-activated cellulose fibers (CF-act) samples were labeled according to the target temperature: (CF-“700–850” and CF-“700–850”-act). Fig. 1 illustrates this novel method for the preparation of porous carbons and the steam-activation process on the carbon surface schematically.

2.2. Characterization

The surface morphology of the CF samples was observed by scanning electron microscopy (SEM, Model SU8010, Hitachi Co., Ltd) at an acceleration voltage of 1 kV. The textural properties of the samples were analyzed with N_2 adsorption–desorption isotherms using a Model Belsorp Max instrument (BEL Japan, Inc.) at 77 K. Prior to analysis, all CF samples were degassed at 200 °C for 10 h. The specific surface area (S_{BET}) was calculated according to the BET (Brunauer–Emmett–Teller) equation with relative pressures (P/P_0) of 0.00003–0.05. The total pore volume (V_{total}) was measured at a relative pressure of 0.99. The micropore volume (V_{micro}) was determined using D–R (Dubinin–Radushkevich) equations, respectively. The pore-size distributions were derived from the BJH (Barrett–Joyner–Halenda) model. The micropore-size distributions were derived from the HK (Horvath–Kawazoe) equation.

2.3. CO_2 adsorption–desorption isotherms

The CO_2 adsorption–desorption isotherms were measured at 298 K and 308 K at low pressure (1 bar) using a Model Belsorp Max instrument (BEL Japan, Inc.). A volumetric measurement method was used to obtain the CO_2 adsorption capacity. Before measurement, the samples were degassed for 10 h at 200 °C until the residual pressure fell below 0.01 kPa and then cooled to room temperature.

2.4. CO_2/N_2 selectivity

To calculate the CO_2/N_2 selectivity of the samples, the CO_2/N_2 adsorption–desorption isotherms were measured at 298 K under low pressure (1 bar) using a Model Belsorp Max instrument (BEL Japan, Inc.).

2.5. CO_2 adsorption–desorption cycle experiment under flue gas condition

CO_2 adsorption–desorption cycle experiments under flue gas conditions were carried out as follows, and were analyzed using TGA (thermogravimetric analysis) method as follows; 5–10 mg of sample was loaded in a platinum pan, and heated to 200 °C at a rate of 5 °C min. This temperature was maintained for 300 min to degas

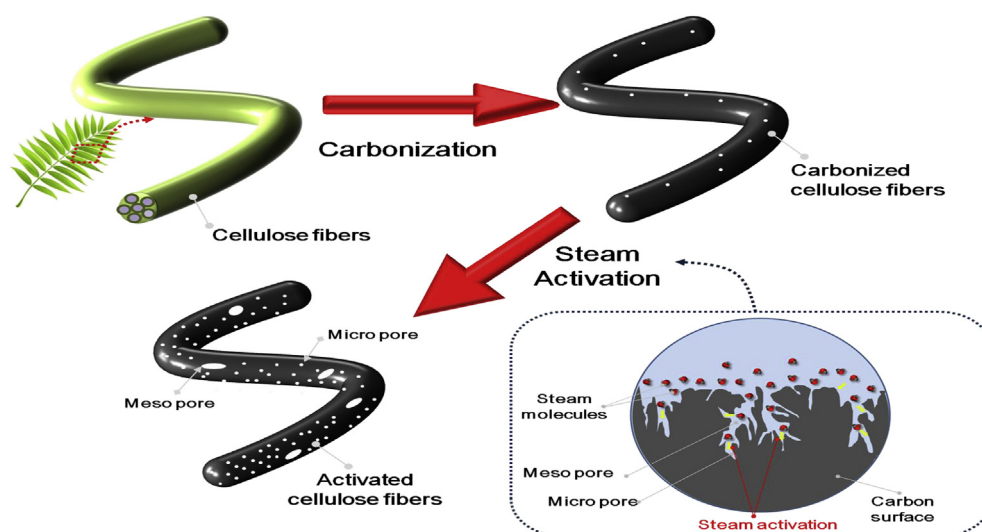


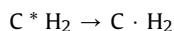
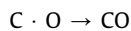
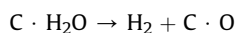
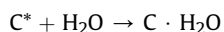
Fig. 1. Schematic diagram of the preparation of porous carbons and steam activation of the carbon surface.

the sample, and then the system cooled to 313 K (5 °C/min) under 100% N₂ gas. After stabilized, the gas flow composition was changed, and the system was maintained at 313 K for 60 min under a gas composition of 15% CO₂ and 85% N₂, which is similar to the composition of flue gas in fossil fuel power plants. There are two types of desorption methods, non-thermal and thermal methods as follows; 1) 15% CO₂ gas flow was changed to N₂ gas and maintained for 60 min to non-thermal desorption process. 2) The system was maintained at 15% CO₂ gas condition, heated to 353 K, and 373 K to thermal desorption process.

3. Results and discussion

3.1. Morphology characterization

Fig. 2 (a–f) shows the SEM images of the pre-activated and subsequently steam-activated CF samples. All samples exhibited similar morphologies and the original shape of the cellulose fibers was retained after activation (Fig. S1). After steam activation, however, newly developed pores were present on the sample surfaces (Fig. 2c and e). Increasing the activation temperature led to the generation of new pores and to pore-size expansion (Fig. 2d–f), suggesting that steam molecules were responsible for a pore-opening process at the surface, thus leading to an increase in surface area. The steam-activation mechanism can be expressed as follows [30].



Steam activation was performed under inert conditions and at temperatures over 700 °C to avoid oxygen erosion of the porous carbon. The mechanism of steam activation may cause expansion of already existing ultra-micropores and the development of new ultra-micropores when steam molecules impact with the carbon matrix. However, the activation process could also destroy graphite structures and expand pores to sizes greater than 2 nm, producing mesopores.

3.2. N₂ adsorption isotherms and pore-size distributions

N₂ adsorption–desorption isotherms were measured to study the textural properties of the CF and CF-act samples. Fig. 3 shows the N₂ adsorption–desorption isotherms at 77 K of all samples. The N₂ isotherms were similar and displayed a type-I shape, which is characteristic of microporous materials. At low relative pressures ($P/P_0 < 0.1$), the N₂ adsorption isotherms indicated a sharp adsorption, followed by a relatively open knee before reaching an equilibrium. This shape implies the existence of microporous carbon with a broad range of micropore sizes. The micropore volume of the preparation samples was calculated using the Dubinin–Radushkevich (D–R) equation;

$$W = W_0 \exp \left[-B \left(\frac{T}{\beta} \right) \log^2 \left(\frac{P_0}{P} \right) \right]$$

where W is the amount of the liquid adsorbate at relative pressure of P/P_0 , W_0 the total amount of adsorbate in the micropores, B the adsorbent constant, and β the affinity coefficient (0.34 for N₂). Table 1 lists the textural properties that are indicative of the pore structure of each sample. The CF samples before activation had high specific surface areas (452–540 m² g^{−1}) and total pore volumes (0.198–0.234 cm³ g^{−1}). After steam activation, the specific surface area (599–1018 m² g^{−1}) and total pore volume (0.248–0.429 cm³ g^{−1}) of the samples increased. Steam molecules, therefore, created samples with higher porosity and a wider pore sizes. The specific surface area and total pore volume increased with increasing temperature; this is understandable because the mechanism of pore formation would depend on the temperature used during steam activation.

Fig. 4 shows the micropore (a, b) and mesopore (c, d) size distribution curves of the samples, which were characterized using the Horvath–Kawazoe (HK) and Barrett–Joyner–Halenda (BJH) methods, respectively. In accordance with the classification of pores (IUPAC), pore sizes between 0.7 and 1 nm are called super-micropores and those less than 0.7 nm are called ultra-micropores. As shown in Fig. 4(a) and (b) for the CF and CF-act samples, the micropore-size distributions were unimodal below 0.7 nm. Fig. 4(a) shows that CF-700 exhibited the highest peak in the microdistribution at 0.66 nm, whereas over 700 °C, all CF samples exhibited a peak at 0.64 nm. Fig. 4 (b) shows that the CF-700-act sample exhibited a new peak in the micropore distribution at

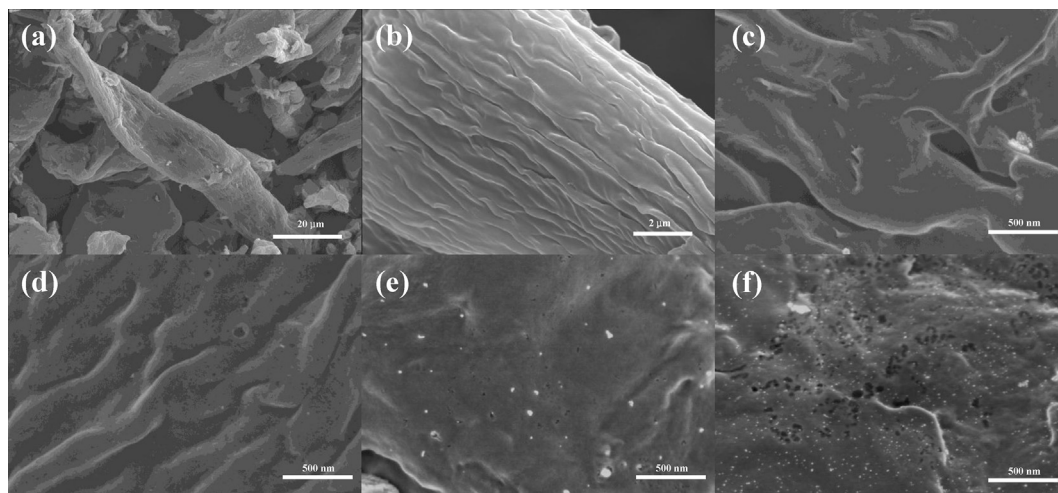


Fig. 2. SEM images of (a–c) CF-800 at various magnifications, (d) CF-800-act, (e) CF-700-act, and (f) CF-850-act.

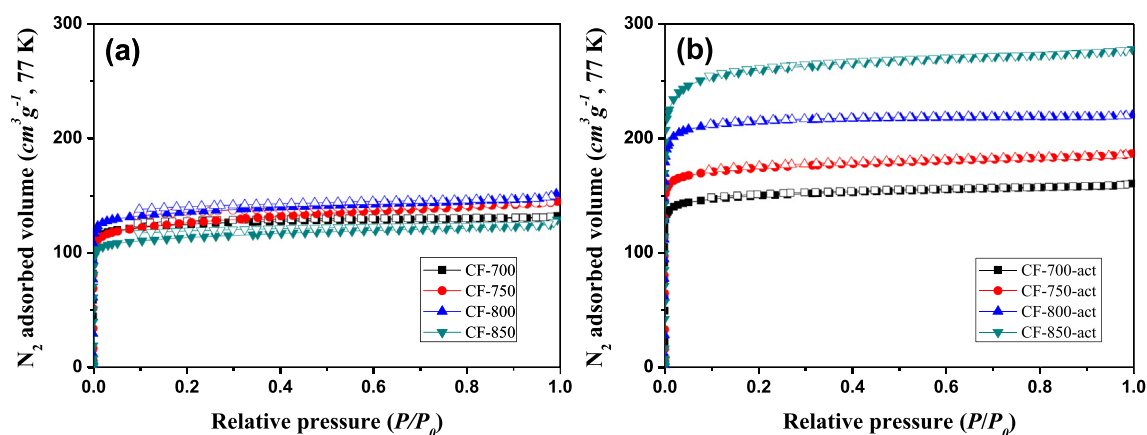


Fig. 3. N₂ adsorption–desorption isotherms of (a) CF and (b) CF-act samples.

0.61, 0.64 nm, whereas the 0.66 nm peak intensity decreased after steam activation. After activation over 700 °C, the peaks exhibited microdistributions of 0.59 and 0.61 nm, whereas the 0.64 nm peak intensity decreased. As shown in the mesopore distributions in Fig. 4(c) and (d), the mesopores mostly consisted of pore sizes less than 30 nm. As shown in Fig. 4(c), as the temperature increased, the intensity of the peak in the mesopore distribution of the CF samples initially increased. It then decreased for temperatures above 750 °C. Meanwhile, the peak intensities in the mesopore distribution curves of the CF-act samples clearly increased with increasing temperature, as shown in Fig. 4(d). This shows that steam activation created new micropores and expanded existing micropores to sizes greater than 2 nm, producing new mesopores. These results demonstrate that the porous structure of the activated samples is dependent on the activation temperature.

3.3. CO₂ adsorption isotherms

The CO₂ adsorption–desorption isotherms of the CF and CF-act samples were measured at 298 K and 1 atm using a pressure swing analysis, as presented in Fig. 5. The CO₂ adsorption capacity is closely related to the specific surface area and micropore volume and size [38]. As shown in Fig. 5, the CF samples had CO₂ uptakes of 97.1–139.9 mg g^{−1} (2.206–3.179 mmol g^{−1}) of adsorbent; on the other hand, the CF-act samples had CO₂ uptakes of 114.8–166.2 mg g^{−1} (2.608–3.776 mmol g^{−1}) at 298 K and 1 bar.

Table 1
Physical characteristics of CF samples.

Specimens	^a S _{BET} (m ² g ^{−1})	^b V _{total} (cm ³ g ^{−1})	^c V _{micro} (cm ³ g ^{−1})	^d V _{meso} (cm ³ g ^{−1})	^e F _{micro} (%)
CF-700	499	0.205	0.193	0.012	94.2
CF-750	494	0.224	0.193	0.031	86.2
CF-800	540	0.234	0.209	0.025	89.4
CF-850	452	0.198	0.174	0.024	88.0
CF-700-act	599	0.248	0.229	0.019	92.5
CF-750-act	696	0.289	0.267	0.022	92.4
CF-800-act	863	0.341	0.334	0.007	98.1
CF-850-act	1018	0.429	0.393	0.036	91.5

^a S_{BET}: Specific surface area computed using BET equation at a relative pressure range of 0.00003–0.05.

^b V_{total}: Total pore volume is estimated at a relative pressure P/P₀ = 0.99.

^c V_{micro}: Micropore volume determined from the D–R equation.

^d V_{meso}: Mesopore volume determined from the subtraction of micropore volume from total pore volume.

^e F_{micro}: Fraction of micropore volume = (micropore volume/total pore volume)*100.

The CO₂ adsorption capacity, therefore, increased with steam activation. The CO₂ adsorption capacity was also observed to improve with increasing steam-activation temperature up to 800 °C. It then decreased at higher temperatures. The highest CO₂ capacity observed, therefore, occurred for a sample activated by steam at 800 °C. At this temperature, the CO₂ uptake increased from 139.9 (for CF-800) to 166.2 mg g^{−1} (for CF-800-act), an improvement in the CO₂ adsorption capacity of 18.6%. At 850 °C, the specific surface area and micropore volume continued to increase; however, the CO₂ adsorption capacity of the CF-850-act sample was observed to be 136.2 mg g^{−1}, which is lower than that of the CF-800-act sample. This indicates that the CO₂ adsorption capacity does not exclusively correlate with the specific surface area and micropore volume; but also depends on the ultra-micropore characteristics. Table 1 shows the relationship between the micropore volume fraction (F_{micro}) and CO₂ adsorption capacity. We can conclude that the CO₂ adsorption capacity was highest for the CF-800-act sample due to their micropore volume fraction and ultra-micropore-size distribution below 0.7 nm, both of which play a key role in improving CO₂ adsorption capacity [19].

3.4. CO₂/N₂ selectivity and the heat of adsorption of CO₂

Henry's law of selectivity (S), calculated based on the Toth model [39] as shown in Equations (a–c), can be used to quantitatively evaluate the selectivity of an adsorbent. To calculate the selectivity of CF-800-act for CO₂ over N₂, CO₂ and N₂ isotherms at 298 K are shown in Fig. 6. The Toth model is given as

$$Q = Q_{\max} A^{(1/n)} P(1 + AP)^{(1/n)} \quad (a)$$

where Q is the gas uptake (mmol g^{−1}), Q_{max} is the maximum gas uptake (mmol g^{−1}), and A and n are fitting constants.

The constant K in Henry's law is given as

$$K = \frac{dN}{dp} = A^{(1/n)} Q_{\max} \quad (b)$$

Henry's law for the selectivity for gas component x over y is given as

$$S_{x,y} = \frac{K_x}{K_y} \quad (c)$$

Based on Equations (a–c), the selectivity of the samples is calculated to be in the range of 28–47. Table 2 clearly shows that all

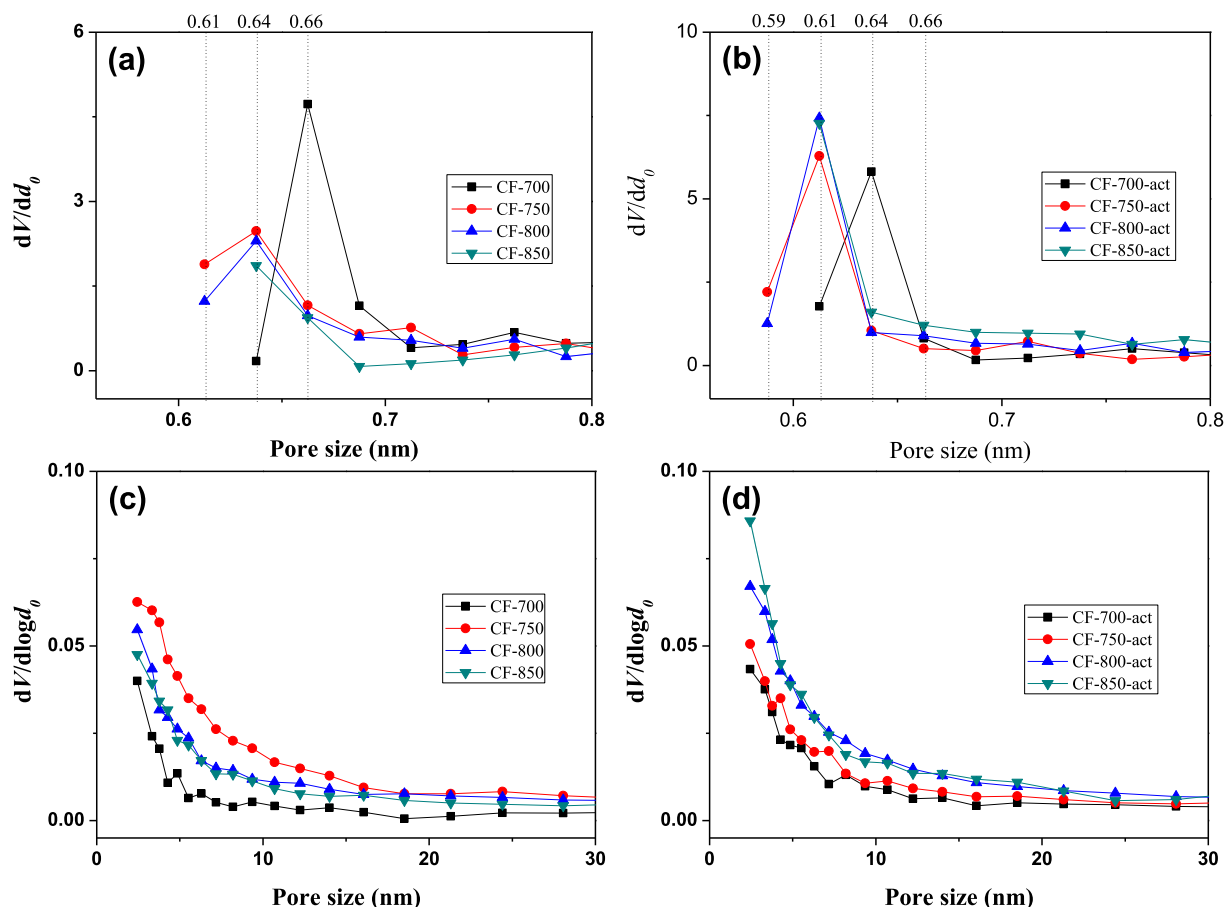


Fig. 4. (a, b) Micro- and (c, d) mesopore size distributions of CF and CF-act samples at various activation temperatures.

the CF-act samples exhibited higher selectivity than the CF samples. The selectivity for CF-800-act in particular was 70% higher than the selectivity for CF-800. Therefore, steam activation led to significant improvement in the CO_2/N_2 selectivity of the CF samples. This improvement in selectivity is likely due to the increased interaction between CO_2 and the adsorbents. The heat of adsorption (ΔH_{ads}) is defined as the amount of heat generated when an adsorbate is adsorbed by an adsorbent, and it is related to the strength of the attractive forces between the adsorbate and the adsorbent. The heat of adsorption of CO_2 can be calculated from the slopes of the CO_2 adsorption–desorption isotherms at 273 K and 298 K using the Clausius–Clapeyron equation [40,41].

$$\ln\left(\frac{P_2}{P_1}\right) = -\frac{\Delta H_{\text{ads}}}{R} \left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

where R is the gas constant, T_1 and T_2 are 273 and 298 K, respectively, and P_1 and P_2 are the equilibrium partial pressures of CO_2 adsorbed at 273 and 298 K, respectively.

Fig. 7 shows the heat of adsorption (ΔH_{ads}) of CO_2 calculated at various CO_2 coverage values. The initial ΔH_{ads} values of the CF-act samples varied in the range of 29–38 kJ mol^{-1} at lower coverage values, and the values were observed to decrease slightly with increasing quantity of adsorbed CO_2 . CF-800-act exhibited the highest value (37.8 kJ mol^{-1}) at lower coverage; however, this

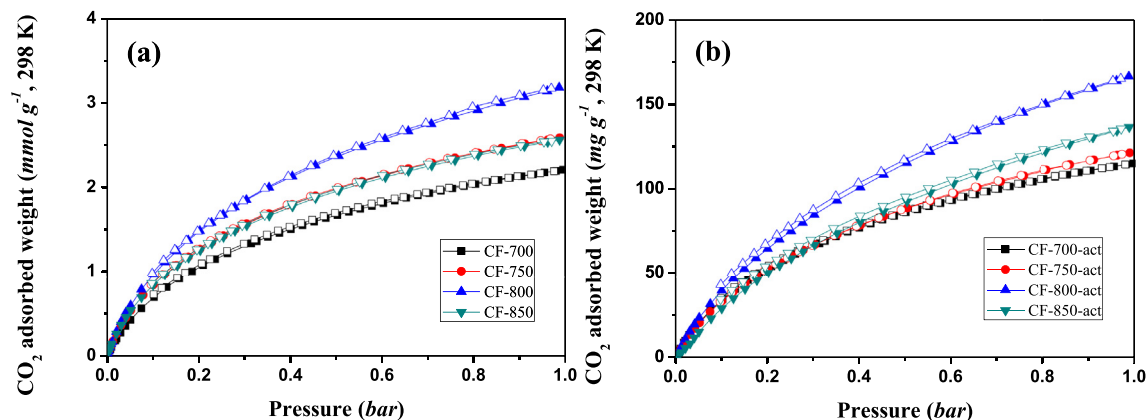


Fig. 5. CO_2 adsorption–desorption isotherms of (a) CF and (b) CF-act samples.

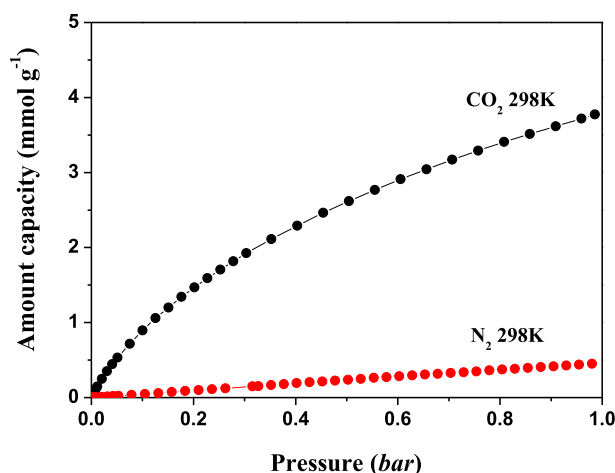


Fig. 6. Adsorption isotherms of CF-800-act at 298 K in N₂ and CO₂.

value decreased with increasing coverage and approached 27.1 kJ mol⁻¹. The results indicate that ultra-micropores were mainly responsible for CO₂ adsorption, probably due to enhanced CO₂ interaction with the adsorbent in these narrow pores. Although these results are lower than those observed for zeolites and MOFs [42,43] and indicate, therefore, that CO₂ is not as strongly bound to porous carbons, the energy required to break the bond between CO₂ and the CF samples for regeneration will, consequently, be much less than that required for zeolites and MOFs, despite their similar loading capacities. All of the values are in the range of physical adsorption, which will enable regeneration of the adsorbent.

3.5. Multi-component CO₂ adsorption–desorption cycle under flue gas condition

The multi-component CO₂ adsorption/regeneration abilities of the prepared samples were measured by TGA method [44,45]. In contrast to single-component adsorption methods, multi-component adsorption method (TGA method) is carried out routinely with and high accuracy for validating real process. Fig. 8 shows the 15% CO₂ adsorption–desorption curves for CF-800 and CF-800-act. After degassing and stabilization under N₂ gas, the CO₂ adsorption process was carried out under flue gas conditions (15% CO₂, 85% N₂). The sample weight was found to increase sharply after the introduction of CO₂, as CO₂ molecules were physically adsorbed onto the carbon surface. After saturation and stabilization, the gaseous environment reverted to 100% N₂ to allow desorption to take place. The results for the 15% CO₂/85% N₂ uptake experiments

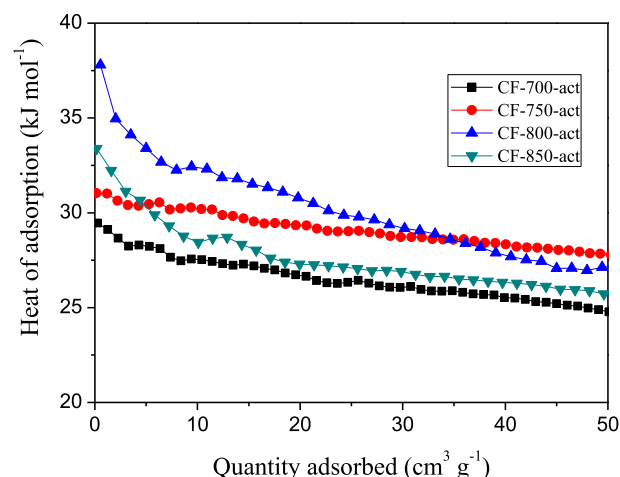


Fig. 7. Heat of adsorption (ΔH_{ads}) of CO₂ on CF-act samples.

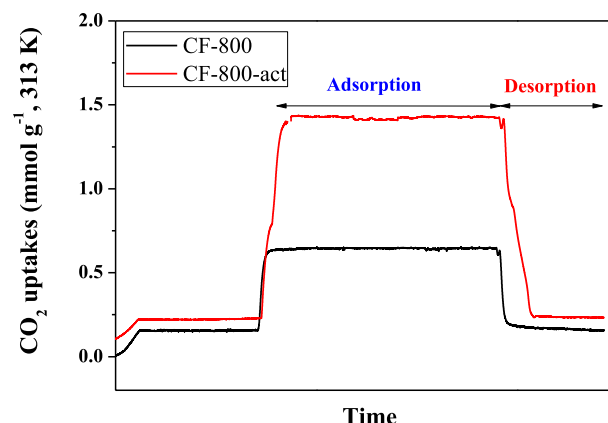


Fig. 8. CO₂ adsorption experiment under flue gas condition for the CF-800 and CF-800-act.

are shown in Table 2. From these results, it can be seen that CO₂ uptake increased from 0.491 to 0.645 to 0.734–1.429 mmol g⁻¹ with steam activation. The CF-800-act sample exhibited the highest CO₂ uptake at 313 K under flue gas conditions, with a CO₂ adsorption of 1.429 mmol g⁻¹.

From an economic point of view, the regeneration properties of CO₂ adsorbents is an extremely important property in assessing the lifetime of the materials in a practical process. The most studies have examined the long cycle regeneration property of adsorbents

Table 2

CO₂ adsorption capacity, CO₂/N₂ selectivity and heat of adsorption for CF samples.

Specimens	CO ₂ adsorption at 1 bar (mmol g ⁻¹)		^a 15% CO ₂ uptakes (mmol g ⁻¹) 313 K	CO ₂ /N ₂ selectivity 298 K	^b ΔH_{ads} (kJ mol ⁻¹)
	273 K	298 K			
CF-700	2.846	2.206	0.491	28.7	27.2–20.1
CF-750	3.271	2.587	0.622	26.4	24.5–21.6
CF-800	3.664	3.179	0.645	27.6	25.1–19.2
CF-850	3.189	2.561	0.551	27.9	25.7–21.0
CF-700-act	3.395	2.608	0.734	35.1	29.5–24.8
CF-750-act	3.792	2.655	0.968	36.7	31.0–27.8
CF-800-act	4.192	3.776	1.429	47.1	37.8–27.2
CF-850-act	4.416	3.096	1.216	39.2	33.4–25.7

^a 15% CO₂ uptakes: CO₂ adsorption capacity was measured at 313 K under flue gas condition.

^b ΔH_{ads} : Heat of adsorption of CO₂ calculated at quantity adsorbed volume range of 0.5–50 cm³ g⁻¹.

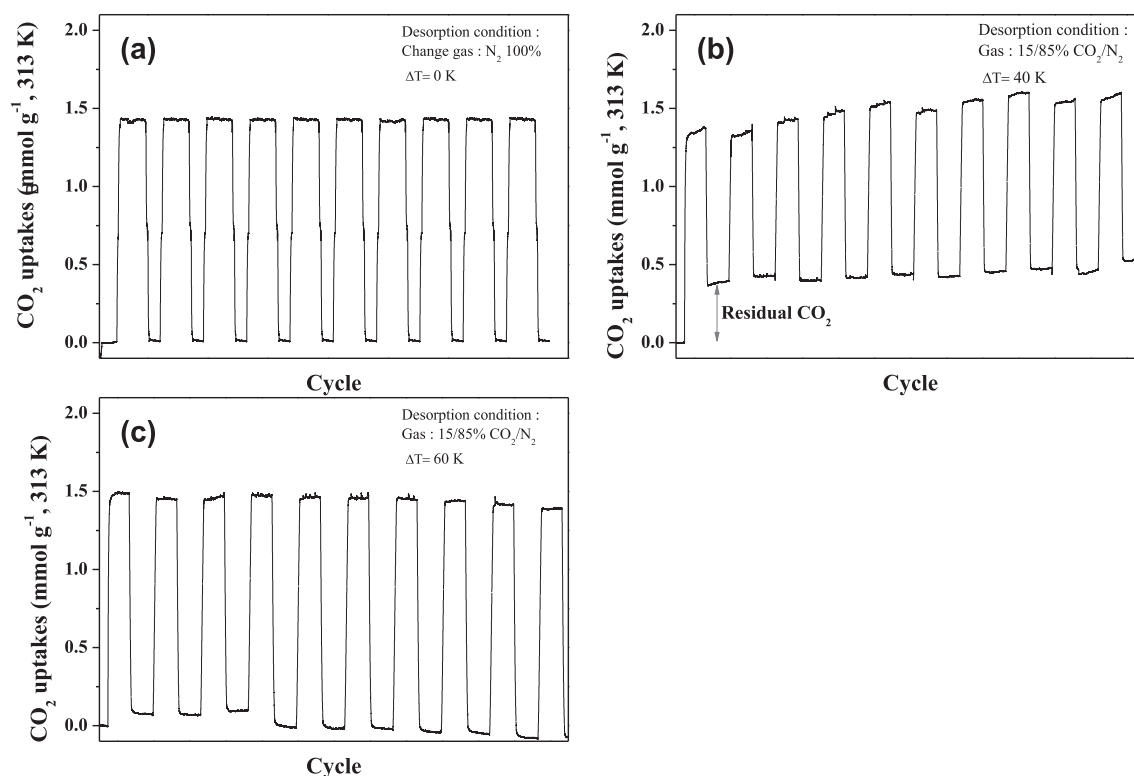


Fig. 9. CO₂ adsorption–desorption cycling tests of CF-800-act under flue gas condition.

through the repeated experiments [45,46]. In this study, we evaluated the regeneration properties using CF-800-act, by cycling the adsorption–desorption process 10 times. The results from these experiments are shown in Fig. 9. As shown in Fig. 9(a), CO₂ desorption process initiated by gas change, CO₂ molecules also easily desorbed at the adsorbents. It is indicated that CO₂ molecules physisorbed onto the CF-800-act. In real application for post-combustion CO₂ capture, we further evaluated CF-800-act using the thermal swing adsorption (TSA) method. In this method, TGA adsorption–desorption cycles were performed by switching between the desorption temperature of 353 K and 373 K and the adsorption temperature of 313 K without switching gases, as noted in the experimental section.

The CF-800-act desorption curves at 353 K and 373 K are displayed in Fig. 9(b) and (c), respectively. The results agree well with the flow system results under flue gas condition. Fig. 9(b) indicates that the adsorbed CO₂ molecules cannot be completely desorbed at

353 K, which might be due to the lack of desorption energy causing incomplete CO₂ desorption from the surface of CF-800-act. On the other hand, Fig. 9(c) shows the adsorbed CO₂ molecules are completely desorbed at 373 K, which could be attributed to adequate desorption energy. Considering all the samples examined in our study, CF-800-act exhibited the highest CO₂ capacity and demonstrated excellent regeneration properties through multiple adsorption–desorption cycles in Fig. 9(a–c). Table 3 summarizes carbon-based adsorbents and their performance in CO₂ capture. As mentioned in this paper, carbon-based adsorbents are considered as one of most promising adsorbents for CO₂ capture. Many researches have shown that the CO₂ capacity of carbon-based adsorbents can be significantly increased by modification of their structures and surface properties. We have shown that CF-800-act can capture CO₂ successfully under flue gas conditions. Thus, we expect that CF-800-act could be a potential adsorbent for industrial CO₂ capture.

Table 3

Summary of carbon-based adsorbents and their performance in CO₂ capture.

Carbon types	Modification	CO ₂ adsorption capacity		Cycle	CO ₂ /N ₂ selectivity	Ref.
		100% CO ₂ (1 bar) (Single-component)	15% CO ₂ (1 bar) (Multi-component)			
Activated carbons	KOH activation	4.18 mmol g ⁻¹ at 298 K	—	0	12	[47]
Activated carbons	Physical activation	2.11 mmol g ⁻¹ at 298 K	—	—	72	[30]
Activated carbons	Steam activation	3.78 mmol g ⁻¹ at 298 K	1.43 mmol g ⁻¹ at 313 K	0	47	Present work
S-doped carbons	S-doped	4.50 mmol g ⁻¹ at 298 K	—	—	51	[48]
N-modified carbons	N-modified	2.07 mmol g ⁻¹ at 298 K	—	0	24	[49]
N-doped carbons	N-doped	2.00 mmol g ⁻¹ at 298 K	—	—	—	[15]
N-doped carbons	N-doped	3.10 mmol g ⁻¹ at 298 K	—	0	38	[50]
Mesoporous carbon nitrides	EDA modified	1.40 mmol g ⁻¹ at 298 K	—	—	—	[51]
Activated carbon fibers	KOH activation	5.68 mmol g ⁻¹ at 298 K	—	—	—	[19]
Activated carbon fibers	Steam activation	2.92 mmol g ⁻¹ at 298 K	—	0	—	[21]
MWCNTs	PEI modified	2.14 mmol g ⁻¹ at 298 K	0.34 mmol g ⁻¹ at 313 K	0	—	[46]

4. Conclusions

Cellulose fiber-based porous carbon adsorbents were prepared by steam activation for enhancing CO₂ capture. Physical activation using steam was effective in developing optimal micropores for CO₂ capture without the need for chemical activation. Steam activation of the carbon also resulted in a significant increase in CO₂/N₂ selectivity due to enhanced interaction energy between CO₂ and the carbon surface. Moreover, high activation temperatures led to the formation of well-developed porosities; this, in combination with the increase in interaction energies, shows that it is possible to control the surface properties of porous carbons through steam activation. These results highlight the importance of porosity for improving CO₂ capture and CO₂/N₂ separation, and demonstrate that steam activation is a promising method for enhancing the CO₂ adsorption capacity of materials and controlling their surface properties. In this work, the CF-800-act exhibits the excellent CO₂ adsorption capacity of 3.776 mmol g⁻¹ at 298 K (1 bar) and good selectivity of 47.1 for CO₂/N₂ separation compared to the performances of carbon-based adsorbents (Table 3). In particular, the CO₂ adsorption/regeneration curves using TSA method for CF-800-act at equilibrium conditions most representative of fossil fuel power plant flue gas condition, revealed the possibility of adsorbing CO₂ gas and regenerating the adsorbent at a lower energy penalty. Thus, CF-800-act represents a promising adsorbent for energy-efficient CO₂ capture and CO₂/N₂ separation from post-combustion process.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.energy.2015.08.033>.

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